

IndyCar Racing 2 - Player Car Aerodynamic Model

Summary of static reverse-engineering findings from the supplied DOS executable

Purpose. This note records the aerodynamic-model findings recovered from the IndyCar Racing 2 executable, with emphasis on the player car. It distinguishes high-confidence findings from interpretations that still require runtime calibration.

Bottom line. The EXE contains a genuine speed-squared aerodynamic model with separate fixed body/underfloor aero and independent front/rear wing terms. Front and rear wing settings are stored in hundredths of a degree, clamped to 3.00-18.00 degrees, and are converted into drag and downforce terms using fixed-point arithmetic.

1. Fixed body / underfloor aerodynamics

Parameter	Road course	Speedway	Interpretation
Body aero coefficient	33,000	28,000	Base body/underfloor drag-related coefficient
Downforce / drag ratio	255	260	Equivalent to 2.55 and 2.60
Rear distribution	43,691	43,691	$43,691 / 65,536 = 0.66667$ rear

The effective internal vertical-aero coefficient is obtained by multiplying the body coefficient by the downforce/drag ratio:

Road: $33,000 \times 2.55 = 84,150$ Speedway: $28,000 \times 2.60 = 72,800$

The body/underfloor contribution is therefore approximately 33.3% front and 66.7% rear. These are relative/internal coefficients; they should not be read directly as pounds or newtons.

2. Wing setup representation

A compact routine in the executable clamps each wing setting between 300 and 1800. The surrounding formatting and setup logic show that the values represent hundredths of a degree:

$300 = 3.00 \text{ deg}$ $1800 = 18.00 \text{ deg}$

Front and rear wing are independent inputs. The code then converts each angle into a drag-related coefficient and an angle-dependent downforce/drag ratio.

3. Reconstructed wing equations

Let $A = 100 \times \alpha$, where α is wing angle in degrees and $300 \leq A \leq 1800$.

Front wing drag coefficient:

$CD_F, \text{int} = 50 + \text{floor}[13,100 \times (A - 300) / 1,500]$

Rear wing drag coefficient:

$CD_R, \text{int} = 100 + \text{floor}[26,200 \times (A - 300) / 1,500]$

Thus, for the same wing angle, the rear wing drag term is approximately exactly twice the front wing term.

The wing efficiency (downforce-to-drag ratio) decreases as angle rises. A compact continuous approximation to the integer routine is:

$L/D \approx 4.5 / [0.5 + (\alpha - 3) / 15]$

This gives approximately $L/D = 9.0$ at 3 deg and $L/D = 3.0$ at 18 deg. Effective wing downforce is then proportional to $CD \times (L/D)$.

Wing angle	Front drag	Front DF	Rear drag	Rear DF	Combined DF
3 deg	50	450	100	900	1,350
5 deg	1,796	12,761	3,593	25,529	38,290
8 deg	4,416	23,846	8,833	47,698	71,544
10 deg	6,163	28,690	12,326	57,380	86,070
12 deg	7,910	32,359	15,820	64,718	97,077
13 deg	8,783	33,877	17,566	67,755	101,632
14 deg	9,656	35,231	19,313	70,466	105,697
16 deg	11,403	37,546	22,806	75,093	112,639
18 deg	13,150	39,450	26,300	78,900	118,350

Note: the table uses the reconstructed continuous L/D approximation and should be treated as representative of the fixed-point integer routine rather than as an independently measured force table.

4. Relative magnitude of body and wings

At maximum wing setting (18 deg front and 18 deg rear), the reconstructed internal vertical-aero terms are:

Component	Internal drag term	L/D	Internal downforce term
Road body / underfloor	33,000	2.55	84,150
Front wing @ 18 deg	13,150	3.00	39,450
Rear wing @ 18 deg	26,300	3.00	78,900
Total	-	-	202,500

For a road-course car at 18 deg / 18 deg, the internal total is therefore divided approximately as follows:

Body / underfloor: $84,150 / 202,500 = 41.6\%$ Front wing: $39,450 / 202,500 = 19.5\%$ Rear wing: $78,900 / 202,500 = 39.0\%$

The maximum combined wing term is about 1.41 times the fixed road-course body/underfloor term. At more moderate wing angles the body contribution becomes proportionally larger.

5. Speed dependence and force architecture

The final code trace shows that these coefficients are applied to a common airspeed-squared term. In simplified form:

Aero term proportional to V_{air}^2 Body drag proportional to $V_{\text{air}}^2 \times CD_{\text{body}}$ Body DF proportional to Body drag $\times R_{\text{body}}$ Front DF proportional to $V_{\text{air}}^2 \times CD_{\text{front}}(\alpha_F) \times R_{\text{front}}(\alpha_F)$ Rear DF proportional to $V_{\text{air}}^2 \times CD_{\text{rear}}(\alpha_R) \times R_{\text{rear}}(\alpha_R)$

Therefore, all else equal, halving airspeed reduces aerodynamic force to one quarter. The code also contains wind speed and direction inputs, which is consistent with the use of relative airspeed rather than only ground speed.

The aero result is passed through the game's general internal force-to-acceleration scaling. A constant of 17,184 appears in that path together with a dynamic vehicle-mass term. This is why the static executable does not support a clean one-step conversion of the internal aero coefficient to pounds or newtons.

6. Why the earlier absolute-force estimates should be treated cautiously

An initial interpretation treated 33,000 / 65,536 as if it were directly an SI $CD \cdot A$ quantity. That produced plausible-looking values (for example roughly 1,400 lb of body downforce at 200 mph), but the deeper code trace

shows Papyrus applies additional internal speed, mass and acceleration scaling. Those absolute force values are therefore not demonstrated by the EXE and should not be presented as exact ICR2 outputs.

The relative coefficients, angle laws, front/rear ratios, speed-squared dependence and body distribution are substantially better supported.

7. Confidence assessment

Finding	Confidence
Wing settings stored as hundredths of a degree	High
Wing range clamped to 3.00-18.00 deg	High
Separate front and rear wing aero terms	High
Rear wing coefficient approximately 2x front at equal angle	High
Wing efficiency decreases from about L/D 9 to 3 across the range	High / reconstructed
Road body coefficient 33,000; speedway 28,000	High
Body downforce/drag ratios 2.55 and 2.60	High
Body aero split approximately 1/3 front, 2/3 rear	High
Aero scales with relative airspeed squared	High
Absolute pounds/newtons from static EXE alone	Not established

8. Practical implementation for a clone

A close behavioural reproduction can be built without knowing the final force-unit conversion. Use one global calibration constant K_{aero} and preserve the recovered ratios:

$$\begin{aligned} DF_{\text{body}} &= K_{\text{aero}} \times V_{\text{air}}^2 \times 84,150 \text{ (road)} & DF_{\text{front}} &= K_{\text{aero}} \times V_{\text{air}}^2 \times \\ CDF_{\text{front}}(\alpha_F) & & DF_{\text{rear}} &= K_{\text{aero}} \times V_{\text{air}}^2 \times CDF_{\text{rear}}(\alpha_R) \end{aligned}$$

Apply approximately 1/3 of fixed body downforce at the front axle and 2/3 at the rear, then add the front-wing term to the front axle and rear-wing term to the rear axle. A single runtime calibration point from the original game could then determine K_{aero} in physical units.

9. Recommended final calibration

To obtain exact pounds/newtons, capture one known in-game aero/load quantity at a known speed and wing setting. Because the coefficient relationships and angle laws are already reconstructed, one trustworthy runtime datum should be enough to establish the force scale for the full model.

Scope note: This summary is based on static examination of the supplied INDYCAR executable and cross-checking of internally repeated parameters. It is intended as a reverse-engineering working note, not an official Papyrus specification.